LINEAR ALGEBRA · LESSON 11

Matrix Spaces; Rank 1; Small World Graphs

Matrices can be vectors too. A covariance matrix lives in one such space, and a single-factor market is a **rank one** matrix.

dim S = n(n+1)/2 | A = uvᵀ | MIT 18.06, Lecture 11

Anything you can add and scale

All 3×3 matrices form a space M of **dimension 9** (basis: 9 matrices with a single 1).

Subspace	Dimension
Symmetric S	6
Upper triangular U	6
Diagonal $D = S \cap U$	3

$\dim S + \dim U = \dim(S + U) + \dim(S \cap U)$**:** $6 + 6 = 9 + 3$**.**

IN MARKETS

Covariance lives in S

A covariance matrix is symmetric. For n assets you need

$$\frac{n(n+1)}{2} \text{ numbers}$$

- 2 assets: 3 (two variances, one covariance)
- 3 assets: 6
- 50 stocks: **1,275**

Diagonal covariance = uncorrelated assets.

RANK ONE

A column times a row

$$\begin{bmatrix} 1 \\ 1.2 \\ 0.5 \end{bmatrix} \begin{bmatrix} -10 & 5 & 20 \end{bmatrix} = \begin{bmatrix} -10 & 5 & 20 \\ -12 & 6 & 24 \\ -5 & 2.5 & 10 \end{bmatrix}$$

Betas × market returns: every row a multiple of one row. **One factor drives everything ⇒ rank 1.**

BUILDING BLOCKS

Rank r = r rank-one pieces

Add a small-cap factor to midcap: $+(0,1,0)^T(4,-2,6)$. Midcap's row becomes $(-8,4,30)$, no longer a multiple of Nifty's.

Two factors ⇒ rank 2.

Rank 1 + rank 1 is usually rank 2, so rank-one matrices are **not** a subspace.

$\mathbf{Cov = \sigma^2 \beta\beta^T + D}$

$$\underbrace{\begin{bmatrix} 400 & 480 & 0 \\ 480 & 576 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{400\,\beta\beta^T \text{ (rank 1)}} + \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 324 & 0 \\ 0 & 0 & 25 \end{bmatrix}}_{D}$$

$$= \begin{bmatrix} 400 & 480 & 0 \\ 480 & 900 & 0 \\ 0 & 0 & 25 \end{bmatrix}$$

50 STOCKS

Full covariance: 1,275 numbers. One factor: 50 + 1 + 50 = **101**.

How far apart are two brokers?

Nodes = brokers, edges = direct trading links.
Distance = fewest links between two nodes.

The adjacency matrix A has a 1 for each link; $(A^2)_{ij}$ counts **two-step** paths.

In finance networks, distance is how many hops a shock takes to spread.

RECAP

Key takeaways

1. Matrices form vector spaces: 3×3 has dim 9.
2. Symmetric: $n(n+1)/2$, where covariance lives.
3. Rank one $= uv^T$: a single-factor model.
4. Rank r = r rank-one pieces = r factors.
5. Graphs: distance, adjacency, A^2.

QUANT FROM FIRST PRINCIPLES

The full interactive lesson is free

A factor builder, a rank one lab, and worked hedging and options examples:

learn.krisarj.com

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